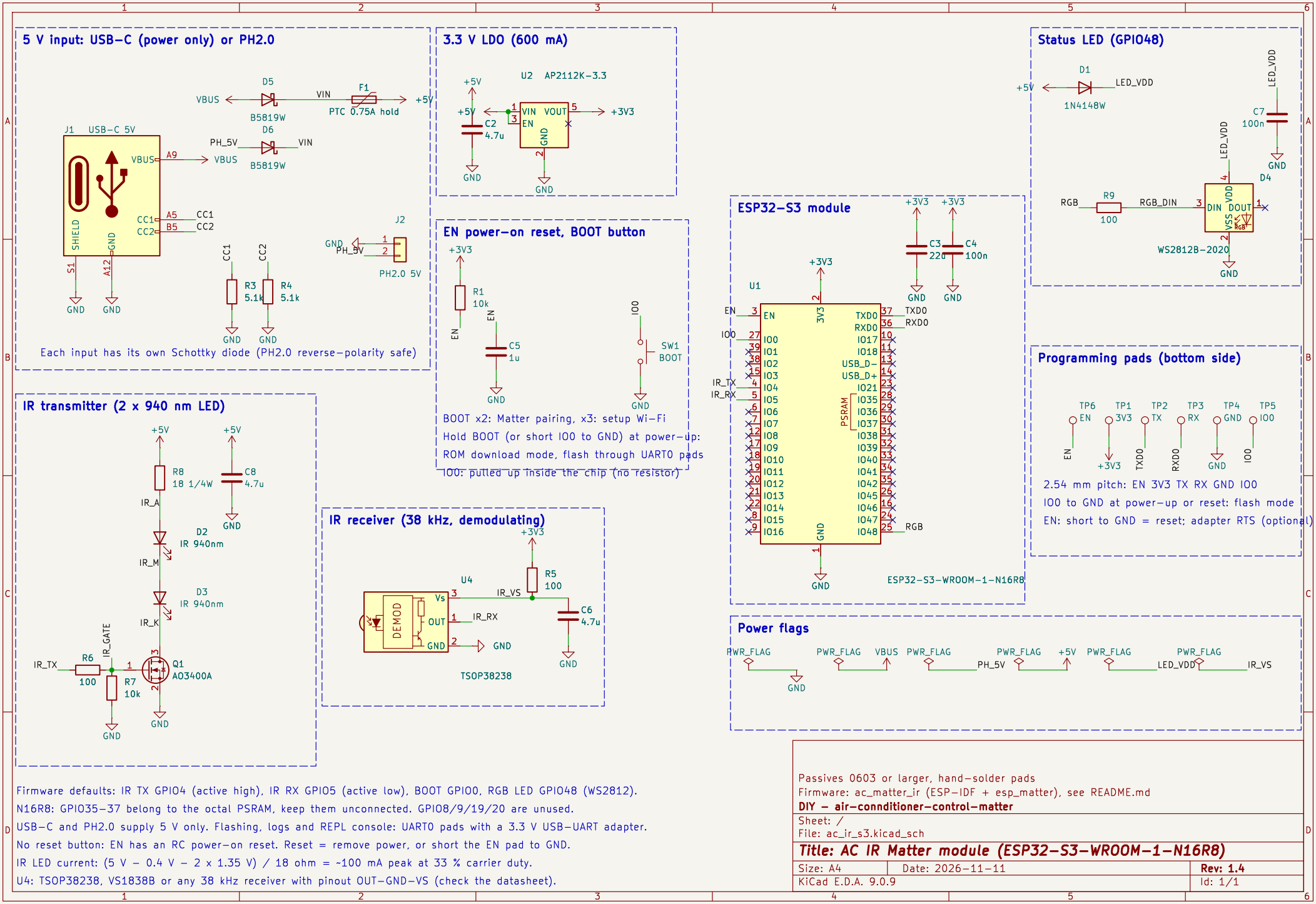

5 V input: USB-C (power only) or PH2.0
J1 USB-C 5V
VBUS
A9
A5 CC1
B5 CC2
SHIELD
GND
A12
S1
D5
B5819W
D6
B5819W
VIN
F1
PTC 0.75A hold
+5V
PH_5V
R3 5.1k
R4 5.1k
J2
PH2.0 5V
Each input has its own Schottky diode (PH2.0 reverse-polarity safe)
3.3 V LDO (600 mA)
U2 AP2112K-3.3
VIN VOUT
EN
GND
C2 4.7u
+3V3
EN power-on reset, BOOT button
R1 10k
C5 1u
SW1 BOOT
IO0
BOOT x2: Matter pairing, x3: setup Wi-Fi
Hold BOOT (or short IO0 to GND) at power-up:
ROM download mode, flash through UART0 pads
IO0: pulled up inside the chip (no resistor)
ESP32-S3 module
U1
EN
IO0
IR_TX
IR_RX
3V3
TXD0
RXD0
PSRAM
RGB
C3 22u
C4 100n
ESP32-S3-WROOM-1-N16R8
Status LED (GPIO48)
D1
1N4148W
LED_VDD
C7 100n
D4
WS2812B-2020
R9 100
RGB_DIN
DIN DOUT
VDD VSS
Programming pads (bottom side)
TP6 EN
TP1 3V3
TP2 TX
TP3 RX
TP4 GND
TP5 IO0
2.54 mm pitch: EN 3V3 TX RX GND IO0
IO0 to GND at power-up or reset: flash mode
EN: short to GND = reset; adapter RTS (optional)
IR transmitter (2 x 940 nm LED)
R8 18 1/4W
C8 4.7u
D2 IR 940nm
D3 IR 940nm
IR_A
IR_M
IR_K
IR_GATE
R6 100
R7 10k
Q1 AO3400A
IR receiver (38 kHz, demodulating)
U4
DEMOD
Vs
OUT
GND
IR_VS
R5 100
C6 4.7u
TSOP38238
Power flags
PWR_FLAG
VBUS
PH_5V
LED_VDD
IR_VS
Firmware defaults: IR TX GPIO4 (active high), IR RX GPIO5 (active low), BOOT GPIO0, RGB LED GPIO48 (WS2812).
N16R8: GPIO35-37 belong to the octal PSRAM, keep them unconnected. GPIO8/9/19/20 are unused.
USB-C and PH2.0 supply 5 V only. Flashing, logs and REPL console: UART0 pads with a 3.3 V USB-UART adapter.
No reset button: EN has an RC power-on reset. Reset = remove power, or short the EN pad to GND.
IR LED current: (5 V - 0.4 V - 2 x 1.35 V) / 18 ohm = ~100 mA peak at 33 % carrier duty.
U4: TSOP38238, VS1838B or any 38 kHz receiver with pinout OUT-GND-VS (check the datasheet).
Passives 0603 or larger, hand-solder pads
Firmware: ac_matter_ir (ESP-IDF + esp_matter), see README.md
DIY – air-connditioner-control-matter
Sheet: /
File: ac_ir_s3.kicad_sch
Title: AC IR Matter module (ESP32-S3-WROOM-1-N16R8)
Size: A4
Date: 2026-11-11
Rev: 1.4
KiCad E.D.A. 9.0.9
Id: 1/1